

SUBRATA PAL

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EDUCATION

Ph.D. Candidate in Organic Chemistry

Department of Chemistry, Oklahoma State University

Aug. 2022 – Present (Expected Dec. 2027)

Stillwater, OK

- Advisor: Dr. Spencer P. Pitre
- Dissertation: "Structure-Activity Relationship Investigation of Cobalt Complexes for Photocatalyzed Reactions Under Visible Light Irradiation." (working title)

MS in Organic Chemistry

Department of Chemistry, Shahjalal University of Science & Technology

Jan. 2016 – Apr. 2018

Sylhet, Bangladesh

- Advisor: Dr. Mohammad Mizanur Rahman Khan
- Dissertation: "Synthesis and Investigation of PVA-ZnS and PVA-AgI-ZnO Nanocomposite Films: Towards Efficient and Stable Photocatalyst for Organic Dye Removal"

B.Sc. in Chemistry

Department of Chemistry, Shahjalal University of Science & Technology

Jan. 2011 – Dec. 2015

Sylhet, Bangladesh

RESEARCH EXPERIENCE

Graduate Research Assistant

Pitre Lab, Oklahoma State University

Jan. 2025 – Present

Stillwater, OK

- Synthesized a library of cobaloxime catalysts and evaluated their catalytic performance alongside commercially available Co(II) catalysts in Giese-type reactions with diverse Michael acceptors and radical precursors to establish structure-activity relationships and identify viable alternatives to Vitamin B₁₂ catalysts.
- Enabled Acid-Free Regioselective Cobalt-Photocatalyzed C-4 Minisci Alkylation through Urea Activation of Pyridines.
- Employed TLC, 1D/2D NMR spectroscopy, and GC-MS to characterize reaction intermediates and products and elucidate catalytic mechanisms.
- Performed computational studies using Gaussian and ORCA for geometry optimization and frequency calculations, followed by energetic evaluations to calculate bond dissociation energies (BDEs), redox potentials, and nucleophilicity of cobalt complexes.
- Mentored one undergraduate researchers.

Graduate Research Assistant

MMR Khan Research Group, Shahjalal University of Science & Technology

Jan. 2016 – Apr. 2018

Sylhet, Bangladesh

- Synthesized PVA-ZnS and PVA-AgI-ZnO nanocomposite films with varied PVA compositions and investigated their photocatalytic activity toward organic dye degradation.
- Employed FT-IR, UV-vis and PL spectroscopy to characterize the nanocomposite films.

PUBLICATIONS

([†] indicates equal contribution & * indicates corresponding author)

2. Pal, S.[†]; Pauze, M.[†]; Shafiei, N.[†]; Pitre, S. P.* "Base Metal Photocatalysts: A Bright Future in Photoredox Catalysis?" *Organometallics* **2026**, 45 (6), 587 – 612.
1. Khan, M. M. R.*; Pal, S.; Hoque, M. M.; Alam, M. R.; Younus, M.; Kobayashi, H. "Simple Fabrication of PVA-ZnS Composite Films with Superior Photocatalytic Performance: Enhanced Luminescence Property, Morphology, and Thermal Stability." *ACS Omega* **2019**, 4 (4), 6144 — 6153.

CONFERENCE PRESENTATIONS

- 67th Annual ACS Pentasectional Meeting. (Apr. 2026) **Pal, S.**; Pauze, M.; Pitre, S. P. "Structure-Activity Relationship Investigation of Cobalt Complexes for Photocatalyzed Reactions Under Visible Light Irradiation", Oral Presentation, Pittsburg State University, Pittsburg, Kansas, USA.
- 66th Annual ACS Pentasectional Meeting. (Mar. 2025) **Pal, S.**; Pitre, S. P. "Investigation of Cobalt(II) Complexes for Photocatalyzed Reactions Under Visible Light Irradiation", Oral Presentation, Oral Roberts University, Tulsa, Oklahoma, USA.
- ACS Southwest Regional Meeting. (Oct. 2024) **Pal, S.**; Pitre, S. P. "Investigation of Cobalt(II) Complexes for Photocatalyzed Reactions Under Visible Light Irradiation", Poster Presentation, Waco Convention Center, Waco, Texas, USA.
- ACS Southwest Regional Meeting. (Nov. 2023) **Pal, S.**; Pitre, S. P. "Cobaloxime-Photocatalyzed Minisci Reactions Under Visible Light Irradiation", Poster Presentation, Omni Oklahoma City Hotel, Oklahoma City, Oklahoma, USA.
- International Conference on Nanotechnology and Condensed Matter Physics. (Jan. 2018) **Pal, S.**; Khan, M. M. R.; Oliullah, M. "Synthesis, Interaction and Thermal Properties of PVA-ZnS Nanocomposite Films: Application as Stable Photocatalyst for Organic Dye Removal", Poster Presentation, Dept. of Physics, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh.

AWARDS & FELLOWSHIP

- **National Science & Technology (NST) Fellowship**, Ministry of Science and Technology, Bangladesh (2017)

TEACHING EXPERIENCE

Graduate Teaching Assistant

Department of Chemistry, Oklahoma State University

Aug. 2022 – Dec. 2024

Stillwater, OK

- Instructed Courses: General Chemistry I Lab, Organic Chemistry I, Survey of Organic Chemistry and Lab.
- Led laboratory sessions of approx. 50 undergraduate students per semester, graded lab reports and exams.

Lecturer (Chemistry)

Green University of Bangladesh (GUB)

Sep. 2019 – Jul. 2022

Dhaka, Bangladesh

- Instructed Courses: General Chemistry, Organic Chemistry (Both Theory & Laboratory).
- Led weekly discussion sections and laboratory sessions for approximately 200 undergraduate students per semester.

LABORATORY & TECHNICAL SKILLS

- **Synthetic Techniques:** Air- and moisture-sensitive (Schlenk/glovebox) techniques, organometallic synthesis, multi-step organic synthesis, photochemical reactions
- **Instrumentation:** NMR (1D/2D), UV-Vis, TLC, GC-MS, FT-IR, Wet chemistry including extractions, distillations, standard/solution preparation
- **Software:** MestReNova, ChemDraw, OriginPro, MS Office Suite, Gaussian, Avogadro, Molden, LaTeX

PROFESSIONAL AFFILIATIONS

- **American Chemical Society (ACS)**, Student Member since Oct. 2025
- **ACS Graduate Student Organization-OSU (ACSGSO-OSU)**, Secretary since Sep. 2025
- **ChemFusion**, Advisor since Mar. 2021

REFERENCES

Dr. Spencer P. Pitre

Associate Professor, Research Advisor

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